

Malaeb Hail Platform

Academic Section: Technologies, System Diagrams, and Use Cases

1. Technologies Used

This section presents the technologies adopted in the Malaeb Hail platform, classified by development domain.

1.1 Web Development

- Python – backend programming language
- Flask – lightweight web framework
- HTML5, Jinja2, JavaScript – interfaces and interaction

1.2 Database

- SQLite (malaeb.db) with parks and bookings tables
- Python sqlite3 module for data access

1.3 UI Design

- CSS3, RTL layout, Cairo font
- Green palette inspired by Hail Municipality branding
- Google Maps embed for location pin

1.4 Version Control

- Git for source code tracking

1.5 Local Server

- Flask Development Server at <http://127.0.0.1:5000>

Category	Technology	Role
Web Development	Python + Flask	Server and routes
Database	SQLite	Store parks and bookings
UI Design	CSS3 + RTL + Maps	Arabic interface and map
Version Control	Git	Version tracking
Local Server	Flask Server	Local runtime

2. Use Cases

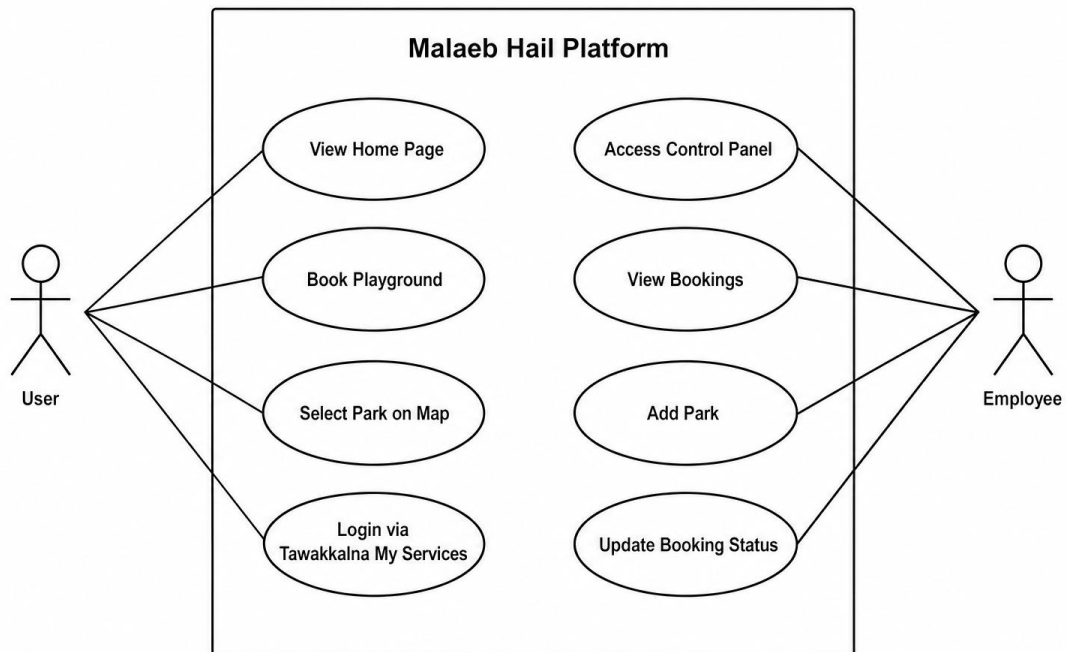


Figure 1. Use Case Diagram

ID	Use Case	Actor
UC01	View Home Page	User
UC02	Book Playground	User
UC03	Select Park on Map	User
UC04	Login via Tawakkalna	User
UC05	Access Dashboard	Employee
UC06	View Bookings	Employee
UC07	Add Park	Employee
UC08	Update Booking Status	Employee

3. System Architecture

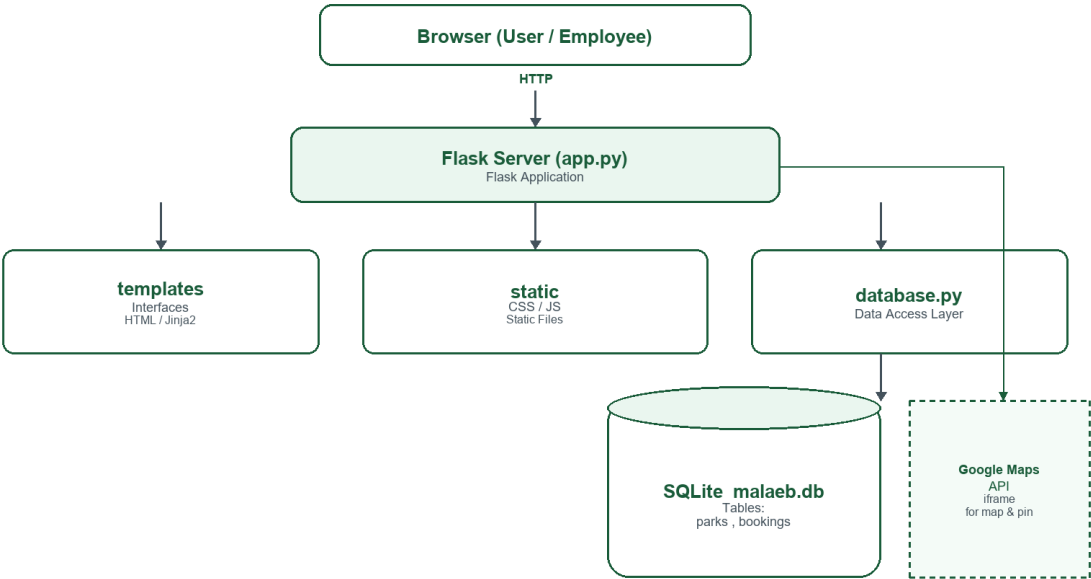


Figure (3): Architectural Diagram for Malaeb Hail Platform

Figure 2. System Architecture Diagram

The platform follows a Client–Server model: Browser → HTTP → Flask → templates/static/database.py → SQLite + Google Maps.

4. Booking Flow

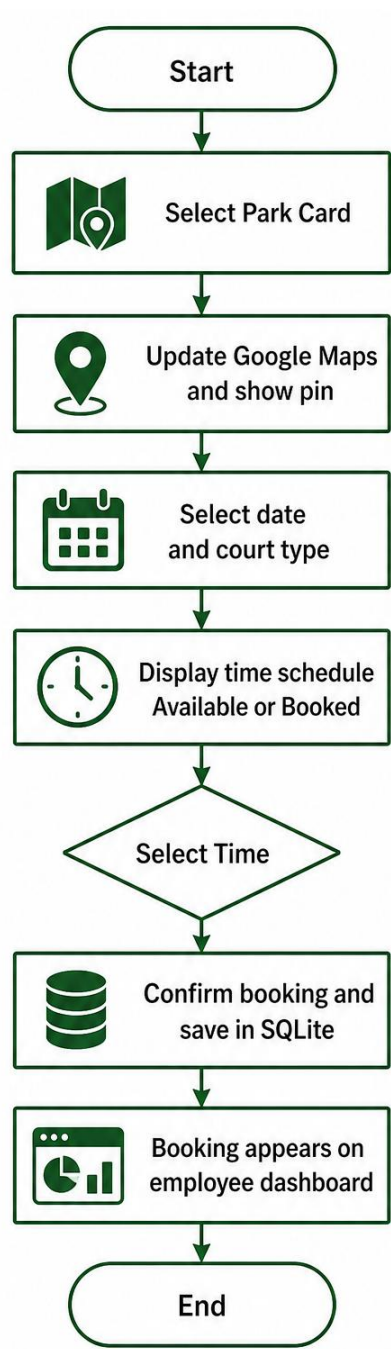


Figure 3. Booking Flow Diagram

The flow covers park selection, map update, schedule selection, booking confirmation, and employee dashboard visibility.

5. Conclusion

The selected technologies and diagrams demonstrate a structured academic design suitable for Smart City municipal sports booking.